

Supermarket Sales — Pandas, NumPy & Data Visualization

100 Practice Questions + Concept Hints

Important rule

Try every question yourself first. Only open the hint when you are genuinely stuck. The hints identify the technique but intentionally avoid giving the full solution.

Core decision rule

If a question says 'for each city/team/product/etc.', think `groupby()`. If it asks how often categories occur, think `value_counts()`. If it says 'show rows where...', think filtering.

Level 1 — Basic Pandas

Section hint: Think about: `shape`, `head()`, `tail()`, `columns`, `dtypes`, `isna().sum()`, `nunique()`.

1. How many rows and columns are there?
2. Display the first 10 rows.
3. Display the last 5 rows.
4. Find the names of all columns.
5. Find the data types of all columns.
6. Check whether there are any missing values.
7. Find the number of unique values in each column.
8. Find the number of unique cities.
9. Find the number of unique product lines.
10. Find the number of unique payment methods.

Level 2 — `value_counts()`

Section hint: Main tool: `df['column'].value_counts()`. For the most/fewest, use `idxmax()/max()` or `idxmin()/min()`.

11. How many transactions came from each city?
12. How many transactions came from each branch?
13. How many transactions were made by each customer type?
14. How many transactions were made by each gender?
15. How many transactions belong to each product line?
16. How many transactions used each payment method?
17. Which product line has the most transactions?
18. Which payment method was used the most?
19. Which city has the most transactions?

20. Which customer type appears the most?

Level 3 — mean(), sum(), max(), min()

Section hint: Select the numerical column first, then apply .mean(), .sum(), .max(), or .min().

- 21. What is the average unit price?
- 22. What is the average quantity purchased?
- 23. What is the average total sales amount?
- 24. What is the maximum total sale?
- 25. What is the minimum total sale?
- 26. What is the total revenue?
- 27. What is the total gross income?
- 28. What is the highest-rated transaction?
- 29. What is the average customer rating?
- 30. What is the total quantity of products sold?

Level 4 — Filtering

Section hint: Use boolean filtering: `df[df['Column'] > value]`. For multiple conditions, use `&` and wrap each condition in parentheses.

- 31. Find all transactions where the quantity is greater than 5.
- 32. Find all transactions where the total is greater than 500.
- 33. Find all transactions from Yangon.
- 34. Find all transactions made by Members.
- 35. Find all transactions where the payment method is Ewallet.
- 36. Find all transactions from the Food and beverages product line.
- 37. Find transactions where the rating is greater than 8.
- 38. Find transactions where quantity > 5 AND total > 500.
- 39. Find transactions from Yangon where the customer is a Member.
- 40. Find transactions where the payment method is Credit card and rating > 7.

Level 5 — groupby()

Section hint: Clue: 'for each'. Pattern: `df.groupby('Category')['NumericalColumn'].aggregation()`.

- 41. Find the total sales for each city.
- 42. Find the average sales for each city.
- 43. Find the total sales for each product line.
- 44. Find the average rating for each product line.
- 45. Find the total quantity sold for each product line.
- 46. Find the average unit price for each product line.
- 47. Find the total gross income for each branch.

- 48. Find the average rating for each branch.
- 49. Find the total sales for each payment method.
- 50. Find the average sales for each customer type.

Level 6 — groupby + maximum/minimum

Section hint: First create a grouped result with `groupby()`, then use `idxmax()/max()` (or `idxmin()/min()`) on that result.

- 51. Which city generated the highest total sales?
- 52. Which product line generated the highest total sales?
- 53. Which branch generated the highest gross income?
- 54. Which product line has the highest average rating?
- 55. Which payment method generated the highest total sales?
- 56. Which customer type has the highest average transaction value?
- 57. Which product line sold the highest total quantity?

Level 7 — Multiple groupby

Section hint: Group by more than one categorical column:
`df.groupby(['Column1','Column2'])['Value'].sum()/mean().`

- 58. Find total sales for each city and product line.
- 59. Find average sales for each city and product line.
- 60. Find total quantity sold by each gender for each product line.
- 61. Find average rating for each customer type and gender.
- 62. Find total sales for each branch and payment method.
- 63. Find average total sales for each city and customer type.
- 64. Find which city + product line combination generated the highest sales.

Level 8 — Sorting

Section hint: Use `sort_values()`. For descending order, `ascending=False`. Then `head(n)` for the top n rows.

- 65. Sort transactions by Total from highest to lowest.
- 66. Find the top 10 transactions by total sales.
- 67. Find the bottom 10 transactions by total sales.
- 68. Sort product lines according to total sales.
- 69. Sort cities according to average sales.
- 70. Find the top 5 transactions with the highest gross income.

Level 9 — Date & Time

Section hint: Use `pd.to_datetime()` first. Then `.dt.year`, `.dt.month`, `.dt.day`, etc. Group dates when needed.

- 71. Convert the Date column into datetime format.

- 72. Find the earliest transaction date.
- 73. Find the latest transaction date.
- 74. Extract the month from the date.
- 75. Find the number of transactions on each date.
- 76. Find total sales for each date.
- 77. Find the date with the highest total sales.
- 78. Find the average sales for each day.

Level 10 — Visualization

Section hint: Think about chart purpose: bar=count/compare categories, histogram=distribution, scatter=relationship, line=trend over time.

- 79. Create a bar chart showing number of transactions by city.
- 80. Create a bar chart showing total sales by product line.
- 81. Create a bar chart showing total sales by branch.
- 82. Create a pie chart showing payment method distribution.
- 83. Create a bar chart showing average rating by product line.
- 84. Create a histogram of Total.
- 85. Create a histogram of Rating.
- 86. Create a scatter plot between Unit price and Total.
- 87. Create a scatter plot between Quantity and Total.
- 88. Create a line chart showing total sales by date.

Level 11 — Mini Analysis

Section hint: Combine earlier techniques. Most will require filtering/groupby + aggregation + sorting or idxmax().

- 89. Which city has the highest average transaction value?
- 90. Which product line generates the most revenue?
- 91. Which product line has the highest average customer rating?
- 92. Do Members or Normal customers have higher average transaction values?
- 93. Which payment method is associated with the highest average transaction value?
- 94. Which city has the highest total gross income?
- 95. Which product line sells the highest quantity but doesn't necessarily generate the highest revenue?
- 96. Find the top 5 individual transactions and display their city, product line, quantity and total.
- 97. Find the city + product line combination with the highest total revenue.
- 98. Find the product line with the highest average unit price.
- 99. Find the branch with the highest average rating.
- 100. Create one visualization that summarizes the sales performance of all product lines.